ARCHITECTURE REFERENCE | V2.0

Sentinel Engine

Entry Scoring Architecture

Multi-Timeframe RSI Cumulative Breakout Detection | Indian Equities

The original single-engine entry scoring system for multi-timeframe RSI recovery detection. Designed for confirmed-trend and long-horizon entry signals with delayed firing. Uses 4 timeframes (4H, Daily, 3-Day, Weekly) across 3 core pools plus structural bonus to produce a maximum entry score of 133.

Bonus Pool: 33 pts | 7 signals | Additive above 100

Pool C (Confirmation): 25 pts | 4 signals | Flat

Pool B (Price Structure): 35 pts | 4 signals | 3 TFs

Pool A (RSI Recovery): 55 pts | 8 signals | 4 TFs

Part of the Scout-Sentinel Dual-Engine Architecture
Scout: Early Detection | Sentinel: Confirmed Trend Entry

System Overview

The Sentinel Engine is the original single-engine entry scoring system designed for multi-timeframe RSI cumulative breakout detection in Indian equities. It represents the complete, unabridged scoring architecture that predates the Scout-Sentinel split. Where the Scout Engine was carved out as a faster, early-detection subsystem optimized for catching recovery signals before price confirms a breakout, the Sentinel Engine retains the full original design with its heavier emphasis on price-structure validation and higher-timeframe confirmation signals.

The Sentinel Engine is purpose-built for **confirmed-trend entry** and **long-horizon positioning**. It fires with deliberate delay relative to the Scout Engine, waiting for multi-timeframe RSI recovery to mature and for price structure to validate the breakout before allocating full position size. This delayed-firing characteristic makes it the preferred engine for swing trades spanning weeks to months, where entry precision matters more than entry speed. The companion Scout Engine may flag a stock days or weeks earlier, but the Sentinel Engine provides higher conviction when it finally triggers.

Architecturally, the system operates on a **trough-gated scoring cycle**. Scoring only begins when a Daily RSI21 trough is detected (RSI21 falls to 40 or below and forms an 11-bar local minimum). From that trough point onward, every subsequent bar within the recovery window is independently scored across three core pools (A, B, C) plus a bonus pool. The core pools are capped at 100 points, and the bonus pool is additive above that cap, yielding a maximum possible score of 133. This cap-plus-bonus structure ensures that strong structural tailwinds (sectoral outperformance, volume surges, secular uptrend corrections) can push a confirmed entry above the standard 100-point ceiling.

The engine evaluates **four timeframes** simultaneously: 4-Hour, Daily, 3-Day, and Weekly. Each timeframe is scored independently, with higher timeframes receiving proportionally higher point values to reflect their greater informational weight and signal rarity. A Weekly RSI divergence is worth four times a 4H divergence. This per-timeframe independence means a signal on any single timeframe contributes its full weight without being suppressed by conditions on unrelated timeframes, enabling the engine to detect multi-spectral recovery patterns that single-TF systems would miss.

Position in the Dual-Engine Architecture

Dimension	Scout Engine	Sentinel Engine
Purpose	Early detection below breakout	Confirmed trend continuation
Firing Speed	Fast (early recovery)	Delayed (confirmed structure)
Pool A (RSI Recovery)	45 pts (Heavy)	55 pts (Heavy)
Pool B (Price Structure)	25 pts (Moderate)	35 pts (Heavy)
Pool C (Confirmation)	15 pts (Light)	25 pts (Significant)
Bonus Pool	25 pts	33 pts

Dimension	Scout Engine	Sentinel Engine
Max Total Score	125	133
Timeframes	3 (D, 3D, W) or 4	4 (4H, D, 3D, W)
Entry Threshold	70+ (Standard)	70+ (Standard)
Typical Hold Period	Days to weeks	Weeks to months
Use Case	Pilot buys, early positions	Full allocation, trend trades

The Scout and Sentinel engines can score the same stock simultaneously. They are not mutually exclusive nor sequential. A stock may appear on the Scout Engine's radar during early RSI recovery, receive a Pilot Buy allocation, and then weeks later trigger a Full Entry on the Sentinel Engine once price structure confirms the trend. This dual-engine approach ensures comprehensive coverage across the entire recovery cycle, from the first flicker of RSI improvement to validated, structure-confirmed trend continuation.

Architecture Overview

The Sentinel Engine's scoring architecture is organized into four distinct scoring layers: three core pools (A, B, C) that are capped at a combined 100 points, plus a bonus pool that adds up to 33 points on top of the core score. This separation between core and bonus ensures that the primary scoring dimensions (RSI recovery, price structure, and confirmation) are evaluated on a normalized 100-point scale, while structural advantages like sectoral outperformance and secular uptrend corrections provide additional conviction without distorting the core evaluation.

Scoring Formula

> **Entry Score = min(100, Pool A + Pool B + Pool C) + Bonus**

The three core pools (A + B + C) sum to a maximum of 115 raw points (55 + 35 + 25), but are hard-capped at 100. This cap ensures that even if all RSI, structure, and confirmation signals fire simultaneously, the core evaluation remains on a standardized scale. The bonus pool (33 points maximum) is then added on top of this capped core, allowing strong structural tailwinds to distinguish exceptional entries from merely good ones. A stock scoring 133 represents the theoretical maximum: full core recovery across all timeframes plus every structural bonus firing simultaneously.

Pool	Classification	Max Pts	Lifecycle	Gate
Pool A	RSI Recovery & Crossovers	55	Dynamic / Live-die	None
Pool B	Price Structure	35	Live / Dynamic	None
Pool C	Confirmation	25	Fire-and-hold / Live	None
Bonus	Structural	33	Live / Dynamic	None
Total		133		None

A critical design decision in the Sentinel Engine is the **absence of inter-pool gating**. Each pool scores independently based on its own conditions. Pool B does not require Pool A signals to activate. Pool C does not require Pool A or B scores to unlock. Grouping into A/B/C is purely organizational: Pool A captures momentum and RSI dynamics, Pool B captures price-structure basing patterns, and Pool C captures higher-conviction confirmation signals. Valid signals from any domain contribute their full weight without being suppressed by unrelated conditions in another domain. This ensures that no genuine recovery signal is missed due to an arbitrary dependency on a different scoring dimension.

Pool A — RSI Recovery & Crossovers

Pool A is the primary scoring engine, carrying the heaviest weight at 55 points. It tracks individual RSI recovery, crossovers, and divergence across all four timeframes (4H, Daily, 3-Day, Weekly). Points grow as RSIs recover from their cycle lows following a detected trough; they shrink if RSIs fall back, reflecting the dynamic and sometimes non-linear nature of real-world recovery patterns. Each of the eight signals is evaluated independently on each timeframe, allowing the pool to capture multi-spectral RSI improvement that would be invisible to single-indicator or single-timeframe systems.

Signal	Condition	4H	D	3D	W	Total	Lifecycle
RSI Divergence	Price LL, RSI HL at local minima	0.5	1.0	1.5	2.0	5.0	Fire-and-hold
RSI21 Reversal	Recovery from cycle low, scaled 0-100%	0-1.5	0-2.5	0-3.0	0-3.5	0-10.5	Dynamic
RSI36 Reversal	Recovery from cycle low, scaled 0-100%	0-0.5	0-1.0	0-1.0	0-1.5	0-4.0	Dynamic
RSI56 Reversal	Recovery from cycle low, scaled 0-100%	0-0.5	0-0.5	0-1.0	0-1.0	0-3.0	Dynamic
RSI21 > RSI36	Bullish crossover (21 crosses above 36)	1.5	2.5	3.0	3.5	10.5	Live/die
RSI21 > RSI56	Bullish crossover (21 crosses above 56)	1.0	1.5	2.0	2.5	7.0	Live/die
RSI36 > RSI56	Bullish crossover (36 crosses above 56)	0.5	1.0	1.5	2.0	5.0	Live/die
All RSI > 50	RSI21, RSI36, RSI56 all above 50	0.5	1.5	2.0	2.5	6.5	Live/die
Max per TF		6.5	12.0	16.0	19.0	53.5	Capped at 55

The raw maximum across all four timeframes sums to 53.5 points, which is then capped at 55. This slight headroom in the cap allows for rounding and accumulation dynamics without penalizing stocks that achieve near-perfect RSI recovery. The 4H timeframe contributes up to 6.5 points, the Daily up to 12, the 3-Day up to 16, and the Weekly up to 19, reflecting the deliberate timeframe weighting where higher timeframes carry approximately 3x the weight of the 4H timeframe.

RSI Reversal Scaling Formula

For each RSI period (RSI21, RSI36, RSI56), the engine tracks the **cycle_low**, defined as the minimum value of that RSI since the most recent trough detection. The recovery ratio is then computed as:

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recovery_ratio = (current_RSI - cycle_low) / (100 - cycle_low)
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Points are assigned using a power-curve scaling function that rewards early recovery more aggressively and asymptotically approaches the maximum as RSI nears 100. The formula is:

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Points = max_pts x min(1.0, recovery_ratio ^ 0.7)
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The exponent of 0.7 creates a concave curve that front-loads point accumulation. At 50% recovery, the engine awards approximately 62% of max points (0.5^0.7 = 0.616). At 80% recovery, it awards 86% (0.8^0.7 = 0.856). This design philosophy ensures that the engine recognizes and rewards meaningful recovery early in the cycle, rather than requiring RSI to approach extreme overbought levels before assigning significant points. If RSI drops below previous values, points decrease proportionally, reflecting the dynamic and reversible nature of the scoring system.

Signal Breakdown

RSI Divergence (5.0 pts max)

Bullish divergence is the highest-conviction single signal in Pool A, awarded when price makes a lower low while RSI makes a higher low at a detected local minimum. This pattern indicates that selling momentum is exhausting even as price continues to decline, a classic precursor to reversal. The signal is fire-and-hold: once triggered at a local minimum, it persists until the next cycle reset, regardless of subsequent RSI movement. The Weekly timeframe awards 2.0 points (40% of the signal's weight) while the 4H awards only 0.5 points (10%), reflecting the significantly higher informational content and rarity of Weekly divergences.

RSI Crossovers (22.5 pts max combined)

Three crossover signals track the progressive bullish alignment of the RSI stack: RSI21 crossing above RSI36 (10.5 pts), RSI21 crossing above RSI56 (7.0 pts), and RSI36 crossing above RSI56 (5.0 pts). These are live/die signals: they activate when the crossover occurs and deactivate if the relationship reverses for a sustained period. The progressive weighting reflects the increasing conviction as shorter-period RSI overtakes longer-period RSI. When all three crossovers are active simultaneously, it indicates full bullish RSI alignment across the stack, a strong confirmation of directional momentum shift.

All RSI Above 50 (6.5 pts max)

This signal fires when all three RSI periods (RSI21, RSI36, RSI56) are simultaneously above 50 on the same timeframe. It represents a broad-based momentum confirmation that the recovery is not confined to a single indicator period. Like the crossovers, it is a live/die signal that activates and deactivates based on current conditions. The Weekly timeframe contributes 2.5 points, making a Weekly "all RSI above 50" a meaningful confirmation event that, combined with other Weekly signals, can significantly boost the total score.

Pool B — Price Structure

Pool B evaluates price-structure signals for evidence of basing and recovery. It carries 35 points and operates across three timeframes (4H, Daily, 3-Day) with a deliberate exclusion of the Weekly timeframe. Weekly price-structure patterns are captured instead by Pool C's confirmation signals, which are better suited for evaluating high-timeframe structural confirmation. Each of the four signals in Pool B fires independently based on its own condition, with no dependency on Pool A scores or signals.

Signal	Check	4H	D	3D	W	Total	Lifecycle
TP Flatten	TP28 range < 10% over 10 bars	2	1	1	0	4	Live
WC Flatten	WC50 range < 10% over 10 bars	2	1	1	0	4	Live
Price > TP	Close > TP28	3	4	6	0	13	Live
TP > WC	TP28 > WC50	3	5	6	0	14	Live
Max per TF		10	11	14	0	35	

Signal Breakdown

TP Flatten & WC Flatten (8 pts combined)

The TP Flatten signal fires when the TP28 (a 28-bar take-profit level derived from recent price action) has ranged less than 10% over the past 10 bars, indicating that the stock is consolidating and forming a base rather than continuing to decline. The WC Flatten signal applies the same logic to the WC50 (a 50-bar weighted close average). Together, these two signals detect the basing pattern that typically precedes a breakout. The 4H timeframe awards 2 points each (4 total) while Daily and 3-Day award 1 point each, reflecting the greater significance of lower-timeframe consolidation as an early warning signal. Both signals are live: they activate when the flattening condition is met and deactivate when the range expands beyond the 10% threshold.

Price Above TP28 (13 pts max)

This is the single highest-weighted signal in Pool B, rewarding stocks whose closing price has moved above their TP28 level. This is a direct breakout confirmation: price has not merely stopped declining but has actively broken above a key structural level. The 3-Day timeframe awards the most points (6) because a 3-Day close above TP28 represents a significant structural breakout, while the 4H awards 3 points for the earlier but noisier signal. This signal is live and deactivates if price falls back below TP28.

TP28 Above WC50 (14 pts max)

The highest-weighted signal in Pool B measures the relationship between the TP28 and WC50 indicators. When TP28 rises above WC50, it indicates that the short-term structural level has overtaken the longer-term weighted average, a classic sign of trend inflection. Like Price Above TP28, this signal is weighted most heavily on the 3-Day timeframe (6 points) and least on the 4H (3 points). The combination of TP > WC and Price > TP on the same timeframe represents a fully confirmed structural breakout on that timeframe, contributing up to 12 points on the 3-Day alone.

Note: The Weekly timeframe has no Pool B signals by design. Weekly price-structure patterns are captured by Pool C's confirmation signals, which are better suited for evaluating high-timeframe structural confirmation with the additional context of divergence and RSI thresholds.

Pool C — Confirmation

Pool C carries 25 points and contains the highest-conviction confirmation signals in the Sentinel Engine. These signals are rarer than Pool A or Pool B signals but carry disproportionate weight when they fire. Unlike the per-timeframe structure of Pools A and B, Pool C signals use flat point values that do not vary by timeframe, reflecting their nature as definitive confirmation events rather than gradual recovery indicators. Each signal fires independently with no dependency on Pool A or B scores.

Signal	Check	Pts	Lifecycle	Notes
W Divergence	Weekly bullish divergence (price LL, RSI HL)	10	Fire-and-hold	Highest single-signal weight
3D Divergence	3-Day bullish divergence (price LL, RSI HL)	5	Fire-and-hold	Confirms W divergence or standalone
D-RSI21 > 60	Daily RSI21 above 60	5	Live/die	Momentum confirmation
Cross-TF Boost	2+ TFs in Pool A with crossover + Price > TP on any	5	Live/die	Multi-TF agreement bonus

Signal Breakdown

Weekly Divergence (10 pts)

At 10 points, the Weekly bullish divergence is the single highest-weighted signal in the entire Sentinel Engine outside of the Bonus pool. It fires when price makes a lower low while Weekly RSI makes a higher low at a detected local minimum. Weekly divergences are extremely rare and carry immense informational weight because they indicate a fundamental shift in the balance of supply and demand at the highest evaluated timeframe. When combined with a 3-Day divergence (5 pts), these two signals alone contribute 15 points to Pool C, representing 60% of the pool's capacity. The fire-and-hold lifecycle means this signal, once triggered, persists until the next cycle reset, providing a persistent structural conviction anchor for the score.

3-Day Divergence (5 pts)

The 3-Day bullish divergence uses the same price-LL/RSI-HL logic as the Weekly divergence but on the 3-Day timeframe. It can confirm a Weekly divergence (adding 5 more points to the 10 already earned) or fire independently as a standalone confirmation signal. The 3-Day timeframe occupies a valuable middle ground between the noise of Daily data and the extreme rarity of Weekly signals, making 3-Day divergence a reliable and moderately frequent confirmation event.

Daily RSI21 Above 60 (5 pts)

This live/die signal activates when the Daily RSI21 exceeds 60, providing direct momentum confirmation that the recovery has reached a meaningful threshold. An RSI21 above 60 indicates not merely that the stock has stopped declining, but that buying momentum has reached a level that historically correlates with sustained advances. This signal deactivates if RSI21 falls back below 60, ensuring that the score accurately reflects current momentum conditions rather than stale historical achievements.

Cross-TF Boost (5 pts)

The Cross-TF Boost is a meta-signal that rewards agreement across multiple timeframes. It fires when two or more timeframes simultaneously have active RSI crossover signals in Pool A, AND price is above TP28 on any single timeframe. This multi-dimensional agreement requirement makes it one of the rarer signals, but when it fires, it indicates that the recovery is not confined to a single timeframe or indicator but represents a broad, multi-spectral shift in the stock's technical profile. The requirement for Price > TP adds a structural validation layer on top of the RSI crossover agreement.

Bonus Pool — Structural Advantage

The Bonus Pool carries 33 points and is structured as two independent sub-pools: Bonus A (Structural, 15 pts) and Bonus B (Uptrend Correction, 18 pts). Unlike the core pools, bonus points are **additive above the 100-point core cap**. This means a stock with a perfect core score of 100 can reach a maximum of 133 when all bonus signals fire. The bonus pool captures structural advantages that exist outside the RSI/price-structure/confirmation framework: volume anomalies, proximity to support, sectoral relative strength, and the high-probability opportunity presented by corrections in secular uptrends.

Bonus A — Structural (15 pts)

Signal	Check	Pts	Lifecycle	Notes
Volume Surge	Volume vs 20-bar avg on Daily TF	5	Live/die	>1.5x=2, >2.5x=4, >3.5x=5
Support Bounce	Proximity to prior structural peak	3	Live/die	<25%=1, <15%=2, <10%=3
Trend Quality	Secular uptrend conviction	10	Live/die	7-component composite

Volume Surge (5 pts)

The Volume Surge signal measures current Daily volume against the 20-bar moving average and awards points on a tiered scale: volume exceeding 1.5x the average earns 2 points, 2.5x earns 4 points, and 3.5x or more earns the maximum 5 points. This signal captures institutional accumulation and panic capitulation events, both of which tend to occur at significant turning points. A volume surge during recovery indicates that the move has participation from large market participants, significantly increasing the probability that the recovery will sustain rather than fade.

Support Bounce (3 pts)

This signal evaluates proximity to a prior structural peak found within a 180-bar lookback from the trough. The reference point is the swing high immediately preceding the highest peak in that window (referred to as the structural peak), not the last completed higher-high pattern top. A gate condition requires that the stock must have fallen at least 15% from that peak before this signal can activate. Points are awarded by proximity: within 25% of the structural peak earns 1 point, within 15% earns 2 points, and within 10% earns the maximum 3 points. This signal captures the high-probability zone where a stock that has fallen significantly is recovering back toward a known structural level.

Trend Quality (10 pts)

The Trend Quality signal is a 7-component composite that evaluates whether the stock is in a genuine secular uptrend with strong fundamentals and structural health. It is the single highest-weighted bonus signal and provides a comprehensive assessment of trend conviction that goes beyond any single indicator.

Component	Check	Pts
Relative Strength (1Y)	Outperforming best-match Nifty sectoral index, 1 year	2
Relative Strength (6M)	Outperforming best-match Nifty sectoral index, 6 months	1
Relative Strength (3M)	Outperforming best-match Nifty sectoral index, 3 months	1
MA Structure	Price above key moving averages	2
Swing Lows Rising	Successive swing lows making higher lows	2
Correction Compression	Current correction smaller than prior	1
Recovery Speed	Price recovering faster than prior correction	1

The Relative Strength components measure the stock's performance against its best-matching Nifty sectoral index rather than a fixed benchmark. A defence stock is compared to Nifty India Defence, an IT stock to Nifty IT, a bank to Nifty Bank, and so on. This sector-relative comparison ensures that performance context is appropriate to the stock's industry, avoiding the distortion that would arise from comparing a cyclical stock to a broad market index during a sector-specific downturn. The 1-year component carries the most weight (2 pts) because sustained sectoral outperformance over a full year is the strongest indicator of genuine structural strength versus temporary momentum.

Bonus B — Uptrend Correction (18 pts)

Bonus B captures one of the highest-probability entry patterns in equity markets: a stock in a confirmed secular uptrend that corrects by 25% or more. These corrections represent institutional accumulation zones where the long-term trend remains intact but short-term price action has created a discounted entry opportunity. Any technical indicator turning positive during such a correction carries extra weight because the structural tailwind of the secular uptrend provides a favorable asymmetric risk-reward backdrop.

Signal	Check	Pts	Lifecycle	Notes
Uptrend Confirmed	Stock in secular uptrend (200-DMA rising, price above)	3	Live	Prerequisite for this sub-pool
Correction Depth	Price fallen 25%+ from recent swing high	5	Live/die	25-35%=3, 35-45%=4, 45%+=5
RSI Turning Positive	Any RSI crossing above 50 from below	3	Live/die	Per-TF: 4H=0.5, D=1.0, 3D=0.5, W=1.0
Price Above Support	Price holding above 180-bar structural peak support	3	Live	Confirms correction is holding

Signal	Check	Pts	Lifecycle	Notes
Volume on Recovery	Above-average volume on up days during recovery	4	Live/die	Confirms institutional buying

All Bonus B signals require "Uptrend Confirmed" as a structural prerequisite. This is not a gate in the traditional sense; other signals (core pools, Bonus A) can still score normally. But the 18 points of Bonus B only accumulate when the uptrend condition is met. The Uptrend Confirmed check requires two conditions: the 200-Day Moving Average must be rising, and the current price must be above the 200-DMA. This dual condition filters for stocks in genuine secular uptrends, excluding stocks that are merely bouncing within a long-term downtrend.

The Correction Depth signal uses a tiered scaling that rewards deeper corrections with more points: a 25-35% correction earns 3 points, 35-45% earns 4 points, and 45% or deeper earns the maximum 5 points. This counterintuitive weighting (deeper correction = more points) reflects the empirical observation that deeper corrections within secular uptrends tend to produce stronger subsequent recoveries, as the correction has flushed out weak holders and created more room for upside before encountering resistance.

Entry Decision Thresholds

The entry decision framework uses total score (core + bonus) to determine both the action and the position size. The thresholds are calibrated for the Sentinel Engine's confirmed-trend profile, where a given score carries higher conviction than the same score on the Scout Engine because the Sentinel Engine's heavier Pool B and Pool C weighting ensures that structural confirmation is well-represented in the total. Position sizing scales with total score including bonus, allowing stocks with exceptional structural tailwinds to receive maximum allocation.

Score Range	Action	Position Size	Rationale
0 - 49	**DON'T ENTER**	0%	Insufficient recovery evidence across all dimensions
50 - 69	**PILOT BUY**	25 - 30%	Early recovery showing promise but lacking full multi-TF confirmation
70 - 79	**STANDARD ENTRY**	60 - 70%	Solid multi-timeframe confirmation with structural validation
80 - 89	**FULL ENTRY**	100%	Strong recovery across all timeframes with bonus tailwinds
90+	**FULL ENTRY (strong)**	100%	Maximum signal agreement plus structural bonus; highest conviction

The Pilot Buy threshold (50-69) allows for early, risk-managed exposure while the remaining signals develop. If the score subsequently rises to 70+, the position is scaled up to Standard or Full Entry. If the score falls back below 50, the Pilot Buy is exited. This graduated approach ensures that the Sentinel Engine's confirmed signals are actionable without requiring all signals to fire simultaneously before any capital is deployed. The difference between the Pilot Buy and the Scout Engine's equivalent is one of conviction: a 60-point score on the Sentinel Engine means more because it requires heavier Pool B and Pool C contribution than the same score on the Scout Engine.

The maximum possible score of 133 (100 core + 33 bonus) is theoretically achievable but extremely rare in practice. It would require all 19 signals across all four timeframes to fire simultaneously, plus every structural and uptrend-correction bonus to activate. In backtesting, scores above 100 are uncommon and represent exceptional entry opportunities where every dimension of the analysis aligns perfectly.

Trough-Gated Architecture

The Sentinel Engine does not score every bar indiscriminately. The entire scoring cycle is initiated and bounded by a **trough detection event** on the Daily timeframe. This trough-gated architecture ensures that the engine only expends computational resources and generates signals during genuine recovery periods, remaining silent during trending, consolidating, or declining phases where entry signals would be premature or misleading.

Parameter	Value	Description
Trough Detection	D-RSI21 <= 40	11-bar local minimum on Daily RSI21
Cycle Reset	On new trough	All recovery baselines reset to new cycle_low
Recovery Window	50 bars	Scoring active for 50 bars after trough detection
Lookback (Support)	180 bars	Structural peak search window from trough
Support Reference	Prior structural peak	Swing high preceding the highest peak, not HH top

When the Daily RSI21 falls to 40 or below and forms an 11-bar local minimum (the lowest RSI21 reading within an 11-bar centered window), a new trough is detected. This event triggers two actions simultaneously: first, all recovery baselines (cycle_low values for each RSI period on each timeframe) are reset to the current readings, establishing the reference point against which all subsequent recovery will be measured. Second, the 50-bar recovery window opens, during which every subsequent bar is independently scored across all pools and timeframes.

The 50-bar recovery window is the operational lifespan of a single scoring cycle. After 50 bars from the trough detection, the engine goes quiet, returning a score of 0 for all subsequent bars until a new trough is detected. This design reflects the empirical observation that meaningful RSI recovery patterns typically manifest within 50 trading days (approximately 2.5 months) of a significant RSI trough. If recovery has not materialized within this window, the stock is either in a structural decline or a prolonged basing pattern that the engine is not designed to capture.

The 180-bar lookback for structural peak identification provides the reference point for the Support Bounce bonus signal. Within this lookback window from the trough, the engine identifies the highest price peak and then finds the swing high immediately preceding it. This preceding swing high becomes the structural support reference against which proximity is measured for the Support Bounce signal. The 180-bar window (approximately 9 months of daily data) provides sufficient history to identify meaningful structural levels while remaining responsive to recent market structure changes.

Signal Lifecycle Types

Every signal in the Sentinel Engine is assigned one of three lifecycle types that govern how it activates, persists, and deactivates. This lifecycle system prevents stale signals from inflating the score and ensures that the total score accurately reflects current market conditions rather than historical achievements. The three lifecycle types are fundamental to the engine's ability to provide real-time, actionable scores that respond dynamically to changing market conditions.

Lifecycle	Behavior	Deactivation	Examples
Dynamic	Points scale continuously with RSI movement; increases as recovery progresses, decreases if RSI falls back	Automatic when RSI drops below previous values	RSI21/36/56 Reversal
Live/die	Binary: active when condition met, inactive when condition fails for a sustained period	Condition no longer met for defined hold period	Crossovers, All RSI > 50, Volume Surge, D-RSI21 > 60
Fire-and-hold	Triggers once at detection event and persists indefinitely until cycle reset	Only on next trough detection (cycle reset)	RSI Divergence, W Divergence, 3D Divergence

The **Dynamic** lifecycle is used exclusively for the RSI Reversal signals (RSI21, RSI36, RSI56). These signals are the most responsive in the engine, with points updating on every bar as the recovery ratio changes. If RSI21 recovers from a cycle_low of 30 to 50, the reversal signal awards approximately 57% of its maximum points. If RSI21 then falls back to 40, the points decrease proportionally. This continuous scaling ensures that the score always reflects the current state of recovery, not just the peak recovery achieved.

The **Live/die** lifecycle applies to the majority of signals in the engine. These signals activate when their condition is met and deactivate when the condition fails for a sustained period. The "sustained" qualifier prevents transient fluctuations from causing rapid on/off toggling. For example, an RSI21 > RSI36 crossover signal will not deactivate on a single bar where RSI21 dips slightly below RSI36; it requires a sustained failure to trigger deactivation. This hysteresis prevents score volatility and ensures that the entry signal remains stable during the decision-making window.

The **Fire-and-hold** lifecycle is reserved for the most significant structural signals: RSI divergences on all timeframes and the higher-timeframe divergences in Pool C. These signals trigger once at the moment of detection (when a local minimum shows price-LL/RSI-HL) and persist for the remainder of the scoring cycle, regardless of subsequent price or RSI movement. The rationale is that a confirmed divergence at a local minimum represents a structural event whose significance does not diminish simply because price continues to move. The signal is only cleared when a new trough is detected and the cycle resets, ensuring a clean slate for each new recovery assessment.

Design Principles

The Sentinel Engine is governed by five foundational design principles that inform every aspect of its architecture, from signal selection to scoring formula to lifecycle management. These principles were established to ensure that the engine produces scores that are interpretable, responsive, and aligned with the empirical realities of multi-timeframe technical analysis in the Indian equity market.

1. No Inter-Pool Gating

Each pool scores independently based on its own conditions. Pool B does not require Pool A signals to activate. Pool C does not require Pool A or B scores to unlock. Grouping into A/B/C is purely organizational: Pool A captures momentum and RSI dynamics, Pool B captures price-structure basing patterns, and Pool C captures higher-conviction confirmation signals. Valid signals from any domain contribute their full weight without being suppressed by unrelated conditions in another domain. This principle ensures comprehensive coverage: a stock with strong RSI recovery (Pool A) but weak price structure (Pool B) still receives appropriate recognition for its momentum improvement, rather than being penalized because an unrelated dimension has not yet confirmed.

2. Per-Timeframe Independence

Each RSI signal (divergence, reversal, crossover) is evaluated independently on each of the four timeframes: 4H, Daily, 3-Day, and Weekly. A bullish crossover on the 4H timeframe scores its full points regardless of whether the Daily or Weekly crossover has occurred. This design reflects the empirical observation that lower timeframes lead higher timeframes in recovery: the 4H RSI will typically show bullish crossover and reversal patterns days or weeks before the Weekly RSI confirms. By scoring each timeframe independently, the engine captures this cascading recovery pattern and can identify stocks in the early stages of multi-timeframe recovery before all timeframes have aligned.

3. Event-Driven Lifecycle

Signals have defined lifecycles that govern how they activate, persist, and deactivate. Dynamic signals (RSI Reversal) scale continuously with RSI movement, providing the most responsive scoring. Live/die signals activate on condition met and deactivate on sustained failure, providing stable binary signals with built-in hysteresis. Fire-and-hold signals (Divergence) trigger once and persist until cycle reset, providing persistent structural anchors. This multi-lifecycle approach prevents stale signals from inflating the score while ensuring that genuinely significant structural events (like Weekly divergences) are not prematurely cleared by short-term noise.

4. Timeframe Weighting

Higher timeframes receive progressively higher point values across all signals. A Weekly RSI divergence (2.0 pts) is worth 4x a 4H divergence (0.5 pts). A 3-Day Price > TP28 (6 pts) is worth 2x a 4H Price > TP28 (3 pts). This weighting reflects two empirical realities: first, higher-timeframe signals are rarer and therefore carry more informational content when they occur; second, higher-timeframe signals are more difficult to manipulate and represent genuine structural shifts rather than intraday noise. The weighting scheme ensures that a stock with strong Weekly signals naturally outscores one with only 4H signals, even if the 4H signals are more numerous.

5. Sectoral Relativity

Relative Strength is measured against the best-matching Nifty sectoral index, not a fixed benchmark like Nifty 50. A defence stock is compared to Nifty India Defence, an IT stock to Nifty IT, a bank to Nifty Bank, and so on. This sector-relative comparison ensures that performance context is appropriate to the stock's industry. During a sector-specific downturn, a stock that declines less than its sectoral peers may still be outperforming on a relative basis, even if it is declining in absolute terms. The sectoral relativity principle ensures that the Trend Quality bonus accurately captures this relative outperformance rather than penalizing stocks for factors outside their control.

End-to-End Scoring Flow

The following describes the complete scoring flow for a single bar, from data ingestion to final score output. Understanding this flow is essential for interpreting backtest results and debugging scoring behavior.

Step 1: Trough Check

The engine checks whether a Daily RSI21 trough has been detected (RSI21 <= 40, 11-bar local minimum). If no trough exists or the last trough was more than 50 bars ago, the bar receives a score of 0 and no further evaluation occurs. The engine remains silent outside the recovery window.

Step 2: Cycle Low Update

For each RSI period (RSI21, RSI36, RSI56) on each timeframe (4H, D, 3D, W), the engine updates the cycle_low to the minimum value observed since the trough detection. This cycle_low serves as the baseline against which all recovery measurements are computed. The cycle_low can only decrease during a scoring cycle; it never increases.

Step 3: Pool A Evaluation

All eight Pool A signals are evaluated across all four timeframes. Divergence signals are checked for fire-and-hold triggers at local minima. Reversal signals compute the recovery ratio and apply the power-curve scaling formula. Crossover signals and the All RSI > 50 signal are evaluated for live/die activation. Raw Pool A points are summed and capped at 55.

Step 4: Pool B Evaluation

All four Pool B signals are evaluated across 4H, Daily, and 3-Day timeframes (Weekly is excluded by design). TP Flatten and WC Flatten check the 10-bar range of their respective indicators. Price > TP28 and TP > WC check the current bar's indicator relationships. Raw Pool B points are summed (max 35).

Step 5: Pool C Evaluation

All four Pool C signals are evaluated with flat point values. Weekly and 3-Day divergence checks run at local minima for fire-and-hold. D-RSI21 > 60 and Cross-TF Boost are evaluated for live/die activation. Raw Pool C points are summed (max 25).

Step 6: Core Score Calculation

The raw sum of Pools A + B + C (max 115) is passed through the min(100, ...) cap to produce the core score. This normalization ensures that the core evaluation remains on a standardized 0-100 scale regardless of how many signals contribute.

Step 7: Bonus Evaluation

Bonus A signals (Volume Surge, Support Bounce, Trend Quality) are evaluated independently. Bonus B signals are evaluated only if the Uptrend Confirmed prerequisite is met. Bonus points are summed separately (max 33) and added to the capped core score.

Step 8: Final Score & Classification

The final score (core capped at 100 + bonus up to 33 = max 133) is compared against the entry decision thresholds. The bar is classified as DON'T ENTER (0-49), PILOT BUY (50-69), STANDARD ENTRY (70-79), FULL ENTRY (80-89), or FULL ENTRY STRONG (90+). Position sizing recommendations are attached based on the classification.

Appendix: Complete Signal Inventory

The following table provides a consolidated inventory of all 23 signals across the four scoring layers, with their point contributions, lifecycle types, and the timeframes on which they operate. This serves as a quick-reference for understanding the full scope of the Sentinel Engine's evaluation capability.

#	Signal	Pool	Max Pts	TFs	Lifecycle
1	**RSI Divergence**	A	5.0	4H, D, 3D, W	Fire-and-hold
2	**RSI21 Reversal**	A	10.5	4H, D, 3D, W	Dynamic
3	**RSI36 Reversal**	A	4.0	4H, D, 3D, W	Dynamic
4	**RSI56 Reversal**	A	3.0	4H, D, 3D, W	Dynamic
5	**RSI21 > RSI36**	A	10.5	4H, D, 3D, W	Live/die
6	**RSI21 > RSI56**	A	7.0	4H, D, 3D, W	Live/die
7	**RSI36 > RSI56**	A	5.0	4H, D, 3D, W	Live/die
8	**All RSI > 50**	A	6.5	4H, D, 3D, W	Live/die
9	**TP Flatten**	B	4	4H, D, 3D	Live
10	**WC Flatten**	B	4	4H, D, 3D	Live
11	**Price > TP28**	B	13	4H, D, 3D	Live
12	**TP28 > WC50**	B	14	4H, D, 3D	Live
13	**W Divergence**	C	10	W	Fire-and-hold
14	**3D Divergence**	C	5	3D	Fire-and-hold
15	**D-RSI21 > 60**	C	5	D	Live/die
16	**Cross-TF Boost**	C	5	Multi	Live/die
17	**Volume Surge**	Bonus A	5	D	Live/die
18	**Support Bounce**	Bonus A	3	D	Live/die
19	**Trend Quality**	Bonus A	10	Multi	Live/die
20	**Uptrend Confirmed**	Bonus B	3	D	Live
21	**Correction Depth**	Bonus B	5	D	Live/die
22	**RSI Turning Positive**	Bonus B	3	4H, D, 3D, W	Live/die
23	**Price Above Support**	Bonus B	3	D	Live
24	**Volume on Recovery**	Bonus B	4	D	Live/die

Total: 24 signals across 4 scoring layers (Pool A: 8, Pool B: 4, Pool C: 4, Bonus: 8). Maximum achievable score: **133 points** (100 core + 33 bonus).